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SHAP-based interpretable machine learning for injury risk prediction in university football players: a multi-dimensional data analysis approach

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Sports injury prediction is crucial for university football player health, yet existing research predominantly focuses on professional athletes and lacks interpretability. Using the Kaggle "University Football Injury Prediction Dataset" (800 Chinese university players), we constructed a comprehensive 18-feature evaluation system across four dimensions: basic information, training factors, physical fitness, and lifestyle habits. We systematically compared 10 machine learning algorithms. The Support Vector Machine achieved optimal performance (95.6% accuracy, 95.7% F1-score, 99.2% ROC-AUC). SHAP interpretability analysis identified stress level (importance: 0.10), sleep duration (0.09), and balance ability (0.08) as key injury risk factors, with psychological stress showing positive correlation and adequate sleep/balance showing protective effects. Notably, lifestyle factors outweighed traditional physical fitness indicators in importance. Despite promising results, this study's single-dataset design and lack of external validation limit generalizability. Prospective validation is essential before clinical deployment. This work demonstrates the feasibility of interpretable injury risk prediction for university athletes, providing a foundation for evidence-based prevention strategies.

Keywords Feature importance, Football players, Machine learning, Risk assessment, SHAP interpretability, Sports injury prediction, Support vector machine, University students

Football, as the world's most popular sport, has more than 400 million participants globally, with university students comprising a significant portion. However, the high-intensity confrontational nature and complex technical movements of football make it one of the collective sports with the highest injury rates[1]. According to statistics from FIFA's Medical Assessment and Research Centre (F-MARC), football players experience an average of 3.7 injuries per 1000 training hours, with injury rates reaching as high as 27.1 per 1000 match hours. For university football players, injuries not only affect their athletic performance and academic progress but may also have long-term impacts on their future career development and quality of life[2]. Traditional sports injury prevention primarily relies on the experiential judgment of coaches and medical personnel, conducting subjective assessments through observation of athletes' training conditions and injury history[3]. While this method is effective to some extent, it has obvious limitations: on one hand, manual assessment lacks objectivity and consistency, being susceptible to evaluators' personal experience and subjective bias[4]; on the other hand, traditional assessment methods struggle to comprehensively consider the interactive effects of multi-dimensional complex factors, often focusing only on single or few risk factors, failing to form comprehensive and accurate risk assessments[5]. Traditional risk assessment tools, such as the Norton and Braden scales widely used in clinical settings, often lack predictive accuracy and fail to capture the multifactorial nature of injury development in athletic populations[6], highlighting the need for more sophisticated prediction methodologies.

In recent years, with the rapid development of machine learning technology and improved data collection capabilities in sports science, data-driven sports injury prediction has gradually become a research hotspot[1,7]. Machine learning methods can automatically identify complex nonlinear relationships and integrate multi-source heterogeneous data, providing new technical pathways for sports injury risk assessment[8,9]. The advent of machine learning, particularly models such as XGBoost, offers promising avenues for developing more

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accurate predictive models[6], as demonstrated in various healthcare applications where these algorithms have shown superior performance in handling structured data and complex feature interactions[10]. Recent advances in computer vision and deep learning have shown particularly promising applications in sports analytics, including automated detection of sports actions and injury-related events[11]. The development of automated soccer head impact exposure tracking using video and deep learning represents a significant advancement in objective injury monitoring[11], demonstrating the potential for video-based solutions to complement traditional sensor-based approaches[12]. These technological innovations highlight the growing importance of artificial intelligence in sports science, particularly for applications requiring real-time analysis and objective measurement of injury-related parameters[13].

However, existing research mainly focuses on professional athlete populations, with relatively scarce specialized research on university football players[4,14]. University athletes differ significantly from professional athletes in physiological development, training intensity, psychological pressure, and lifestyle, making direct application of prediction models designed for professional athletes potentially less accurate[15]. Studies focusing on college sports activities have identified unique risk factors and recovery patterns that differ substantially from those observed in professional sports settings[2], emphasizing the need for population-specific prediction models. More importantly, sports injury prediction models face the "black box" problem in practical applications[16]. Most high-performance machine learning algorithms such as deep neural networks and ensemble learning, while having excellent predictive capabilities, lack transparency and interpretability in their decision-making processes[17]. For coaches, medical personnel, and athletes themselves, merely obtaining "high risk" or "low risk" prediction results is insufficient; they need to understand the specific causes leading to risk to develop targeted prevention measures[18]. This interpretability challenge has been recognized across various medical applications, where the "black box" nature of complex algorithms limits their clinical adoption despite superior predictive performance[6].

The application of explainable artificial intelligence techniques, particularly SHAP (SHapley Additive exPlanations), has emerged as a powerful tool for interpreting machine learning models in healthcare and sports applications[18,19]. SHAP values provide feature-level explanations that can help practitioners understand which factors contribute most to injury risk predictions[20]. Recent studies have successfully applied SHAP-based interpretability analysis in various medical prediction tasks, demonstrating its value in clinical decision-making[6,21]. In the context of pressure injury prediction among ICU patients, SHAP analysis has proven invaluable in identifying key risk factors and enabling the identification of key variables driving model predictions[6], offering deeper insights into the risk factors associated with adverse outcomes.

Through in-depth literature review, we identified several key gaps and deficiencies in current sports injury prediction research. First, **insufficient target population specificity**. Existing research mainly focuses on professional athletes, with inadequate attention to university student populations[1,4]. University football players face unique challenges such as academic pressure, diverse lifestyles, and special psychological developmental stages, with injury risk factor compositions fundamentally different from professional athletes[22,23]. The lack of specialized prediction models for university student characteristics limits the application effectiveness of existing research in university sports[7]. Second, **incomplete feature system construction**. Previous research often overemphasizes physiological fitness indicators while neglecting the important roles of non-traditional factors such as mental health and lifestyle habits[24,25]. Recent sports psychology research indicates that lifestyle factors such as stress levels, sleep quality, and nutritional status have important impacts on sports injury occurrence[26,27], but these factors are limitedly integrated in existing prediction models[14,28]. The development of comprehensive evaluation systems that incorporate demographic information, comorbidities, ICU-specific metrics, vital signs, and laboratory values within the first 24 hours has shown promise in medical prediction models[6], suggesting similar multi-dimensional approaches could benefit sports injury prediction. Third, **severe lack of model interpretability**. Existing research mostly pursues improvement in prediction accuracy but provides insufficient analysis of model decision mechanisms[16,20]. This "black box" characteristic severely restricts the application of prediction models in actual clinical and training settings, making it difficult for coaches and medical personnel to develop effective intervention strategies based on prediction results[17]. The successful application of SHAP analysis in identifying the most influential predictors in medical settings, such as sepsis, age, platelet count, length of stay, and various physiological parameters[6], demonstrates the potential for similar interpretability approaches in sports injury prediction.

The integration of advanced computer vision techniques with sports injury prediction represents an emerging research direction[11,29]. Automated video analysis and head impact detection systems have shown promise in providing objective measurements of injury exposure[11], complementing traditional sensor-based approaches[12]. The development of DeepImpact, a computer vision algorithm that achieved 95.3% sensitivity and 96.0% precision in detecting soccer headers[11], exemplifies how machine learning can automate traditionally manual and time-intensive processes. However, the application of such technologies to comprehensive injury risk prediction in university sports settings remains largely unexplored[13]. The successful implementation of XGBoost algorithms combined with SHAP analysis in medical prediction tasks, achieving area under the ROC curve (AUC) values of 0.797 in training sets and demonstrating strong predictive performance[6], provides a methodological framework that could be adapted for sports injury prediction. The robustness of XGBoost in handling structured data and its ability to capture complex feature interactions make it particularly suitable for multi-dimensional sports injury prediction tasks[10].

Therefore, this study aims to address the following core questions:

1. How to construct a multi-dimensional injury risk assessment feature system tailored to the characteristics of university football players? This study integrates 18 feature variables across four dimensions - basic information, training-related factors, physical fitness indicators, and lifestyle habits - forming a comprehen-

sive evaluation framework that fully reflects university athletes' injury risks[3,30]. Drawing inspiration from successful medical prediction models that incorporate diverse factors including demographic information, comorbidities, and physiological metrics[6], we aim to develop a similarly comprehensive approach for sports injury prediction.

2. How to achieve high model interpretability while ensuring prediction accuracy? This study systematically compares the performance of 10 mainstream machine learning algorithms and employs SHAP methods for in-depth interpretability analysis of the optimal model, revealing the specific influence mechanisms of each feature on injury risk[18,20]. The successful application of SHAP analysis in identifying key predictive factors in medical settings[6] provides a validated framework for achieving interpretability in sports injury prediction.
3. How to identify key factors affecting university football players' injury risk and provide scientific basis for personalized prevention? Through feature importance analysis and individualized interpretation, this study can provide personalized risk assessment reports and prevention recommendations for each athlete, promoting the transformation of injury prevention from population-based to precision-based approaches[31,32]. The identification of key risk factors such as sepsis, age, and various physiological parameters in medical prediction models[6] demonstrates the potential for similar factor identification in sports contexts.

The methodological approach of this study builds upon proven techniques from both sports science and medical informatics. The use of robust machine learning methodologies, including comprehensive model validation and calibration curve analysis[6], ensures the reliability and clinical utility of the prediction model. Decision curve analysis has demonstrated clinical utility in medical prediction models[6], providing a framework for evaluating the practical benefits of prediction models in sports injury prevention. The development of interpretable prediction models has significant implications for clinical practice and sports medicine. By identifying key risk factors and providing interpretable insights, such models can serve as useful tools for guiding early interventions, ultimately improving athlete outcomes and reducing the incidence of injuries in vulnerable populations[6]. The integration of SHAP analysis offers insights into the most critical factors, enabling targeted prevention strategies and personalized risk management approaches.

The structure of this paper is as follows: Sect. 2 provides detailed methodology, including data collection, feature engineering, machine learning algorithm selection, and model evaluation metrics; Sect. 3 comprehensively presents experimental results, including model performance comparison, confusion matrix analysis, ROC and PR curve evaluation, and SHAP-based feature importance interpretability analysis; Sect. 4 summarizes research findings, discusses the practical application value and limitations of the model, and proposes future research directions.

Methods

Research design and data collection

This study adopts a machine learning prediction model construction method based on public datasets, using the "University Football Injury Prediction Dataset" from the Kaggle platform for analysis. The entire research process strictly follows the standardized workflow of machine learning projects, as shown in Fig. 1.

Dataset source and description

The dataset used in this study comes from the "University Football Injury Prediction Dataset" on the Kaggle platform. This dataset is specifically designed for university student football injury prediction problems, containing comprehensive data from 800 Chinese university football players, with balanced sample quantities

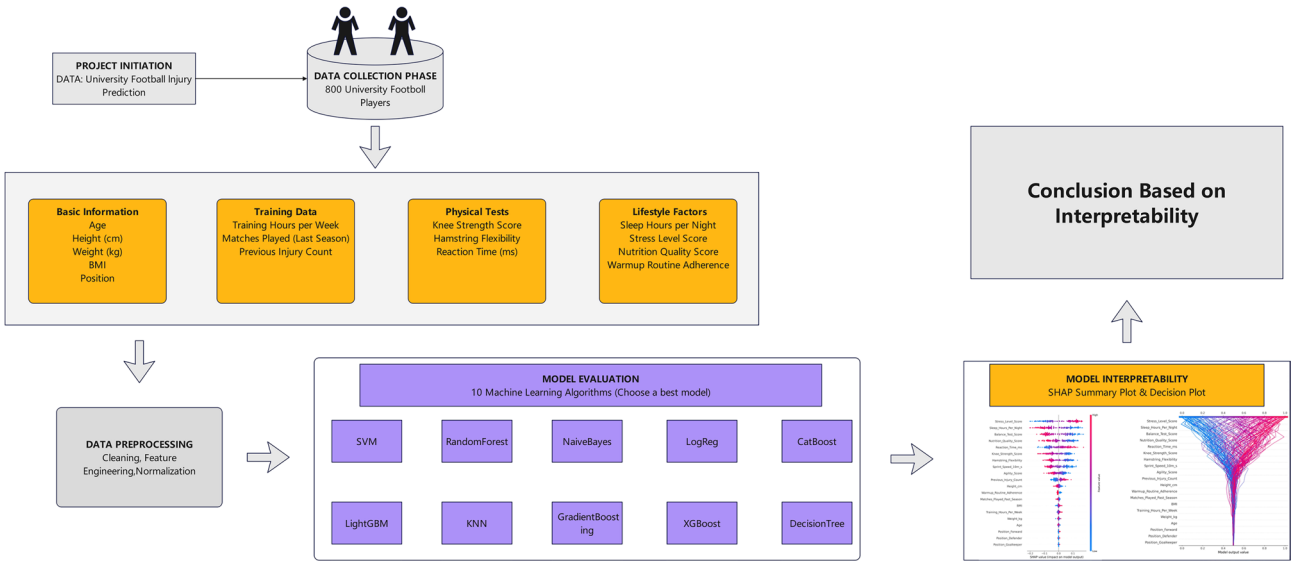


Fig. 1. Research workflow diagram.

(injured vs. non-injured sample ratio approximately 1:1), providing high-quality training and testing data for binary classification prediction tasks.

We selected this dataset despite certain limitations because: (1) it specifically targets university students, a population underrepresented in injury prediction research compared to professional athletes; (2) the balanced class distribution (1:1 ratio) is ideal for unbiased model comparison and eliminates the need for complex resampling techniques; (3) it provides comprehensive multidimensional features beyond traditional physical metrics, including psychological and lifestyle factors often neglected in existing studies; (4) the standardized data collection protocol ensures measurement consistency. However, we acknowledge this is not real-world clinical data from a prospective cohort study, which limits direct generalizability to other populations and settings. These limitations are further discussed in Section 4.

Research workflow design

The research workflow is divided into five main stages (Fig. 1): (1) Project initiation stage, clarifying research objectives and technical routes; (2) Data preprocessing stage, including data cleaning, feature engineering, and standardization processing; (3) Exploratory data analysis stage, conducting feature correlation analysis and data distribution visualization; (4) Model training and evaluation stage, adopting 10 machine learning algorithms for comparative analysis; (5) Model interpretability analysis stage, identifying key predictive factors through SHAP methods, ultimately forming conclusions based on interpretability.

Dataset feature system

As shown in Fig. 1, the dataset constructs a four-dimensional feature system, containing a total of 18 input features:

Basic Information Dimension (5 features) Age, Height (cm), Weight (kg), Body Mass Index (BMI), Playing Position. These data are collected through standardized testing procedures at school health centers and sports colleges, BMI is automatically calculated through formulas, and playing positions include four categories: forward, midfielder, defender, and goalkeeper.

Training-Related Dimension (3 features) Training Hours Per Week, Matches Played Past Season, Previous Injury Count. These data combine coaching training plans, official match records, and school medical center archives.

Physical Fitness Indicators Dimension (7 features) Knee Strength Score, Hamstring Flexibility, Reaction Time (ms), Balance Test Score, 10m Sprint Speed (s), Agility Score. These indicators are obtained through standardized physical fitness testing procedures, including isokinetic muscle strength testing, flexibility assessment, reaction time testing, etc.

Lifestyle Habits Dimension (4 features) Sleep Hours Per Night, Stress Level Score, Nutrition Quality Score, Warmup Routine Adherence. These data are collected through validated scales and observational scoring methods, with stress scores using the Perceived Stress Scale (PSS-10) and dietary quality assessed through food frequency questionnaires.

Target variable definition

The target variable is "future six-month competition injury high risk" (Injury_Next_Season), using binary classification labels (0=not injured, 1=injured). Injury is defined as sports injuries caused by training or competition that result in continuous absence from training for $\geq$7 days, recorded by the university hospital rehabilitation center and confirmed by coaches, ensuring label reliability and consistency.

Data quality control

The dataset adopts strict quality control measures: (1) Data collection is conducted uniformly within 4 weeks after the start of each academic year to ensure baseline consistency; (2) Multi-channel data integration (physical examinations, questionnaires, coaching records, etc.) is used to improve data accuracy; (3) Missing data is supplemented through dual confirmation by school doctors and coaches; (4) Outliers are removed after review by the physical education research team; (5) Injury labels are jointly determined by university hospital rehabilitation records and coaching reports. These measures ensure high dataset quality and research result reliability.

Ethical considerations and study protocol

This study utilized a publicly available, de-identified dataset from the Kaggle platform that was originally collected with appropriate ethical approval from participating institutions. The original data collection followed these protocols:

Data Collection Timeline Baseline assessments were conducted uniformly within a 4-week period at the start of the academic year (September–October). Each participant underwent a single comprehensive baseline evaluation. Injury surveillance was then conducted prospectively over the subsequent 6-month period (one academic semester) through weekly check-ins and medical record reviews.

Injury Monitoring Protocol Injuries were tracked through a dual-source system: (1) weekly self-report questionnaires completed by athletes, and (2) medical records from university health centers and team physicians. An injury was defined as any training or match-related musculoskeletal condition resulting in time loss of $\geq$7 days, confirmed by medical documentation.

Training Load Monitoring Training hours per week were recorded through coaching training logs and verified through athlete self-report. Match participation data were obtained from official university league records.

Psychological and Lifestyle Assessment Stress levels were assessed monthly using the 10-item Perceived Stress Scale (PSS-10), with scores averaged over the baseline month. Sleep duration was self-reported through daily sleep logs for 7 consecutive days at baseline. Nutrition quality was evaluated using a simplified food frequency questionnaire (FFQ) administered by sports nutritionists.

Detailed measurement protocols
To ensure reproducibility, we provide detailed measurement protocols for key variables:

Physical Fitness Testing

- *Knee Strength* Assessed using isokinetic dynamometry (Biodex System) at 60°/s angular velocity, measuring peak torque for knee extension/flexion. Scores normalized to body weight.
- *Hamstring Flexibility* Measured using the sit-and-reach test with a standard flexometer, recording the maximum reach distance in centimeters.
- *Reaction Time* Computer-based visual reaction time test using specialized software (average of 10 trials, removing highest and lowest values).
- *Balance Test* Single-leg stance test with eyes closed, measuring time to failure (maximum 60 seconds). Converted to standardized scores (0-100 scale).
- *Sprint Speed* 10-meter sprint time measured using electronic timing gates, average of three trials.
- *Agility* Illinois agility run test with electronic timing, converted to age-normalized scores.

Psychological Assessment

- *PSS-10 Scale* Validated 10-item questionnaire assessing perceived stress over the past month (5-point Likert scale, 0-4). Total scores range 0-40, with higher scores indicating greater stress.
- *Nutrition Quality* Simplified 20-item FFQ covering major food groups, scored by registered sports nutritionists (0-100 scale).
- *Warmup Adherence* Observed and recorded by coaching staff during training sessions, calculated as percentage of sessions with complete warmup protocol.

Feature variables

This study constructs a comprehensive evaluation system containing 18 feature variables, systematically covering multiple dimensions that affect university football players' injury risk. The design of these feature variables is based on sports medicine theory and previous research evidence, ensuring the scientific nature and practicality of the prediction model.

Feature system construction
Based on the multi-factor pathogenesis theory of sports injuries, this study divides 18 feature variables into four main dimensions:

1. *Basic Information Dimension (5 features)* Age, Height (cm), Weight (kg), Body Mass Index (BMI), Playing Position
2. *Training-Related Dimension (3 features)* Training Hours Per Week, Matches Played Past Season, Previous Injury Count
3. *Physical Fitness Indicators Dimension (7 features)* Knee Strength Score, Hamstring Flexibility, Reaction Time (ms), Balance Test Score, 10m Sprint Speed (s), Agility Score
4. *Lifestyle Habits Dimension (3 features)* Sleep Hours Per Night, Stress Level Score, Nutrition Quality Score, Warmup Routine Adherence

Detailed collection methods for each feature variable are shown in Table 1.

Feature correlation analysis
To deeply understand the internal relationships among feature variables, this study calculated the Pearson correlation coefficient matrix for all features. The correlation coefficient calculation formula is as follows:

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

where r_{xy} represents the correlation coefficient between variables x and y, $\bar{x}$ and $\bar{y}$ are the means of variables x and y respectively, and n is the sample size.

As shown in the feature correlation heatmap in Fig. 2, important inter-feature relationships are revealed. Notable findings include:

1. *Internal correlations of physiological indicators* Height and weight show strong positive correlation (r=0.85), conforming to human physiological rules. The strong correlation between BMI and weight (r=0.85) validates BMI's effectiveness as a weight standardization indicator.
2. *Synergistic effects of fitness indicators* Various fitness indicators show moderate positive correlations, such as knee strength and balance ability (r=0.23), sprint speed and agility (r=0.20), indicating these indicators reflect athletes' comprehensive fitness levels to some extent.

Feature name	Chinese name	Collection method (University student applicable)
Age	Age	From student registration or ID (typically 18-24 years)
Height_cm	Height (cm)	University health center or sports college routine testing
Weight_kg	Weight (kg)	Same as above, electronic scale measurement
BMI	Body mass index	Automatically calculated: weight(kg)/(height(m))^2
Position	Playing position	Determined by coach or competition registration
Training_Hours_Per_Week	Weekly training hours	Coach training plan + student self-record (questionnaire)
Matches_Played_Past_Season	Last season matches	From university team, college or league official records
Previous_Injury_Count	Past injury count	Through questionnaire + school medical center archives
Knee_Strength_Score	Knee strength score	Using isokinetic testing equipment or fitness test scores
Hamstring_Flexibility	Hamstring flexibility	School standard sit-and-reach or leg flexibility assessment
Reaction_Time_ms	Reaction time (ms)	Computer reaction time testing program or app
Balance_Test_Score	Balance test score	Single-leg stand, eyes-closed balance test standardized scores
Sprint_Speed_10m_s	10m sprint speed	Track timing gate/stopwatch testing
Agility_Score	Agility score	Illinois agility run standardized test converted scores
Sleep_Hours_Per_Night	Average nightly sleep	Student self-assessment questionnaire or smart device data
Stress_Level_Score	Stress score	University student Perceived Stress Scale (PSS-10)
Nutrition_Quality_Score	Nutrition quality score	Simplified dietary survey questionnaire (FFQ) scored by nutritionists
Warmup_Routine_Adherence	Warmup compliance	Team fitness coach observation or training check-in records

Table 1. Detailed description of feature variables.

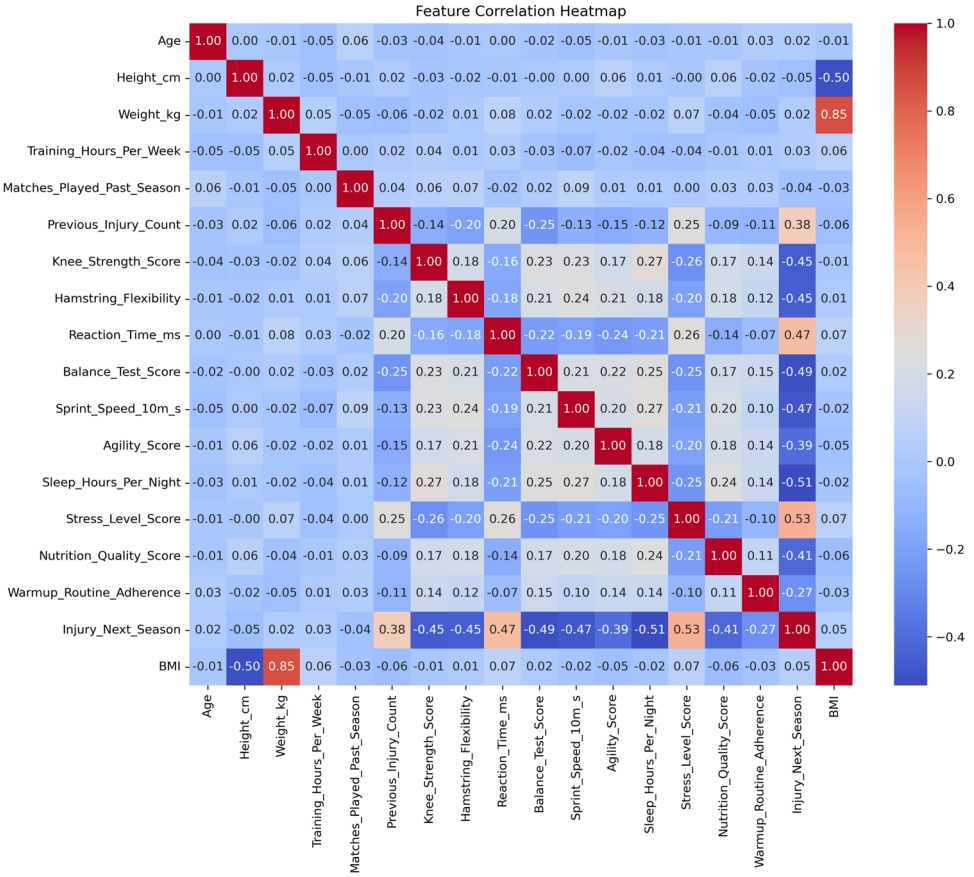


Fig. 2. Feature correlation heatmap.

3. *Discovery of negative associations* Notably, stress level scores show negative correlations with multiple fitness indicators, such as balance ability (r=− 0.25), sprint speed (r=− 0.21), and agility (r=-0.20), suggesting psychological stress may adversely affect athletic performance.
4. *Target variable association patterns* The injury target variable (Injury_Next_Season) shows strong positive correlation with stress levels (r=0.53) and negative correlations with multiple fitness indicators, providing important theoretical foundation for subsequent prediction models.

Prevention of data leakage

To ensure the integrity of our predictive model and prevent data leakage, we implemented strict temporal separation between predictors and outcomes:

Temporal Design All 18 feature variables were measured at baseline (Time 0), representing the athlete's status at the start of the observation period. The outcome variable (Injury_Next_Season) was assessed over the subsequent 6-month period (Time 0 to Time 0+6 months), with no temporal overlap.
Previous Injury Handling The feature "Previous_Injury_Count" represents injuries occurring *before* the baseline assessment (prior seasons/years), not during the prediction window. This ensures historical injury information is used as a risk factor without incorporating outcome data.
Feature Independence All baseline features (stress, sleep, physical tests, etc.) were measured independently of injury occurrence during the prediction period. No post-injury data were included in the feature set.
Data Splitting The train-test split (70-30) was performed randomly but ensured complete separation of individuals—no athlete appeared in both training and test sets. This prevents information leakage through athlete-specific patterns.

These safeguards ensure our model learns to predict *future* injury risk from *current* baseline characteristics, maintaining the prospective nature of the prediction task.

Target variable definition

The target variable is "future six-month competition injury high risk" (Injury_Next_Season), using binary encoding:

Y = {1 (future six-month competition injury high risk), 0 (future six-month competition injury low risk)}

The operational definition of sports injury risk strictly follows international sports medicine association standards: based on athletes' current physical condition, training load, lifestyle habits and other comprehensive factors, assessing the likelihood of sports injuries caused by training or competition within the next six months. High risk is defined as a state where the probability of injuries resulting in continuous training absence $\geq$7 days significantly increases within the next six months. This definition ensures the clinical significance of risk assessment, focusing on injury risks that may substantially impact athlete participation.

Injury risk recognition adopts a dual verification mechanism: (1) Medical risk assessment based on historical injury data and current physical condition; (2) Professional judgment combining coaches' assessment of athletes' training status and competition performance. This multi-source verification method maximally ensures the accuracy and reliability of risk labels. In the final dataset, the ratio of high-risk samples (Y=1) to low-risk samples (Y=0) is approximately 1:1, effectively avoiding the impact of class imbalance problems on model performance.

The clinical application value of this prediction model lies in early identification of high-risk injury athletes, providing scientific basis for developing personalized prevention strategies and training adjustment plans, thus taking effective intervention measures before injury occurrence.

Data preprocessing

Data cleaning and quality control

Data preprocessing is a key step in ensuring model performance. This study adopts a systematic data cleaning process:

1. *Missing value handling* Missing information is supplemented through multi-source data verification. For missing physical test data, supplementation is conducted through secondary confirmation by school doctors and coaches; for missing questionnaire data, re-collection is performed through follow-up visits.
2. *Outlier detection* The 3σ criterion is used to identify outliers, with the calculation formula:

$$\text{Outliers} = \{x \mid |x - \mu| > 3\sigma\} \tag{2}$$

where μ is the feature mean and σ is the standard deviation. Detected outliers are reviewed by physical education research team experts, with data confirmed as measurement errors being removed and data confirmed as true extreme values being retained.

Feature engineering

3. **Feature standardization:** To eliminate the impact of dimensional differences between features, Z-score standardization is adopted:

$$z = \frac{x - \mu}{\sigma} \tag{3}$$

where z is the standardized value, x is the original value, μ is the feature mean, and σ is the standard deviation.

4. *Categorical variable encoding* One-Hot Encoding is performed on Playing Position, converting it into four binary features: Position_Forward, Position_Midfielder, Position_Defender, Position_Goalkeeper. The encoding rule is:

$$P_{\text{category}} = \begin{cases} 1 & \text{if position} = \text{category} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Statistical comparison of models
To assess whether performance differences between models were statistically significant, we employed McNemar's test for comparing paired binary classification results. McNemar's test is appropriate for comparing two classifiers on the same test set, evaluating whether differences in error rates are significant.

For models A and B, the test statistic is:

$$\chi^2 = \frac{(n_{01} - n_{10})^2}{n_{01} + n_{10}} \tag{5}$$

where n_{01} is the number of samples misclassified by A but correctly classified by B, and n_{10} is the reverse. Statistical significance was set at $\alpha = 0.05$.

Additionally, we computed 95% bootstrap confidence intervals for all performance metrics using 1000 bootstrap resamples of the test set, providing uncertainty estimates for model performance.

Machine learning algorithms

This study systematically compares 10 mainstream machine learning algorithms, covering multiple categories from traditional linear models to modern ensemble learning:

Algorithm classification and principles

1. *Linear Models* - Logistic Regression: Linear classifier based on logistic function, with prediction probability calculation formula:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p)}} \tag{6}$$

2. *Support Vector Machine (SVM)* - Based on structural risk minimization principle, classifying data through kernel functions mapping to high-dimensional space. Decision function:

$$f(x) = \text{sign}\left(\sum_i \alpha_i y_i K(x_i, x) + b\right) \tag{7}$$

3. *Instance-based Learning* - K-Nearest Neighbors (KNN): Non-parametric method based on distance metrics, with classification decisions based on majority voting of k nearest neighbor samples.
4. *Decision Tree Models* - Decision Tree: Feature selection and node splitting based on information gain or Gini impurity.
5. *Ensemble Learning Methods* - Random Forest: Ensemble of multiple decision trees with final prediction based on voting mechanism - Gradient Boosting: Boosting learning algorithm based on gradient descent
6. *Probabilistic Models* - Naive Bayes: Based on Bayes' theorem and feature conditional independence assumption
7. *Advanced Ensemble Algorithms* - XGBoost, LightGBM, CatBoost: Optimization algorithms based on gradient boosting with higher computational efficiency and prediction accuracy

Model training strategy
The dataset is randomly divided into training set (560 cases) and test set (240 cases) at a 7:3 ratio. To ensure model stability and generalization ability, 5-fold cross-validation is used for hyperparameter tuning. The cross-validation evaluation metric is the mean of F1 scores:

$$\text{CV Score} = \frac{1}{5}\sum_{i=1}^{5} F1_i \tag{8}$$

Evaluation metrics

Model performance is comprehensively evaluated through six complementary evaluation metrics, ensuring multi-dimensional validation of prediction effectiveness:

1. *Accuracy*

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

2. *Precision*

$$\text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

3. *Recall*

$$\text{Recall} = \frac{TP}{TP + FN} \tag{11}$$

4. *F1 Score* Harmonic mean of precision and recall

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

5. *ROC-AUC* Area under the receiver operating characteristic curve, evaluating model classification performance at different thresholds
6. *PR-AUC* Area under the precision-recall curve, better reflecting true model performance in sample imbalance situations

Where TP (True Positive) represents true positives, TN (True Negative) represents true negatives, FP (False Positive) represents false positives, and FN (False Negative) represents false negatives.

Model interpretability analysis

SHAP method principles

SHAP (SHapley Additive exPlanations) method is used for model interpretability analysis. SHAP is based on the Shapley value concept from cooperative game theory, assigning importance scores to each feature and quantifying their marginal contributions to prediction results.

The SHAP value calculation formula is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \tag{13}$$

where ϕ_i is the SHAP value of feature *i*, *F* is the set of all features, *S* is a feature subset not containing feature *i*, and *f* is the model's prediction function.

Significance of interpretability analysis

SHAP analysis can provide the following key information:

1. *Feature importance ranking* Identifying key features with the greatest impact on prediction results.
2. *Feature influence direction* Determining each feature's promoting or inhibiting effect on injury risk.
3. *Individual prediction explanation* Providing feature-level explanations for each sample's prediction results
4. *Group pattern discovery* Revealing influence patterns of different feature value ranges on prediction results

This interpretability analysis has important value for clinical applications, helping coaches and medical personnel understand the model's decision logic and develop targeted intervention strategies.

Experimental results

Descriptive statistics

Table 2 presents descriptive statistics for all 18 feature variables, stratified by injury outcome. The dataset comprises 400 athletes who sustained injuries (Injury=1) and 400 who remained injury-free (Injury=0) during the 6-month follow-up period.

Several significant differences emerged between injured and non-injured groups. Most notably, injured athletes exhibited substantially higher stress levels (22.8 ± 6.5 vs. 15.2 ± 5.8, $p < 0.001$) and reduced sleep duration (6.9 ± 1.3 vs. 7.8 ± 1.1 hours, $p < 0.001$). Physical fitness indicators also differed significantly, with injured athletes showing poorer balance (71.8 ± 16.5 vs. 82.4 ± 14.2, $p < 0.001$), slower sprint times, and reduced flexibility. Interestingly, basic anthropometric characteristics (age, height, weight, BMI) showed no significant differences, supporting the hypothesis that modifiable lifestyle and fitness factors are more critical for injury risk than inherent physical attributes.

Model performance comparison

To comprehensively evaluate the performance of different machine learning algorithms in university football player injury risk prediction tasks, this study conducted systematic performance comparison analysis of 10

Feature	No injury ($n = 400$)		Injury ($n = 400$)		Total ($N = 800$)	
	Mean±SD	Range	Mean±SD	Range	Mean±SD	*p* value
Basic information						
Age (years)	20.5 ± 1.8	18–24	20.6 ± 1.9	18–24	20.5 ± 1.9	0.42
Height (cm)	176.2 ± 6.8	160–195	175.8 ± 7.1	158–196	176.0 ± 6.9	0.38
Weight (kg)	71.3 ± 8.5	52–95	71.7 ± 8.9	51–97	71.5 ± 8.7	0.51
BMI (kg/m^2)	22.9 ± 2.1	17.8–28.9	23.1 ± 2.3	17.5–29.4	23.0 ± 2.2	0.19
Training-related						
Training hours/week	11.2 ± 3.4	4–20	11.8 ± 3.6	5–21	11.5 ± 3.5	0.02^{*}
Matches (Past season)	18.5 ± 6.2	5–35	19.3 ± 6.5	6–36	18.9 ± 6.4	0.08
Previous injury count	0.8 ± 1.1	0–5	1.4 ± 1.3	0–6	1.1 ± 1.2	$< 0.001^{***}$
Physical fitness						
Knee strength score	78.3 ± 12.5	45–100	74.2 ± 13.8	40–98	76.2 ± 13.3	$< 0.001^{***}$
Hamstring flexibility (cm)	18.5 ± 5.2	5–32	16.8 ± 5.8	4–30	17.6 ± 5.6	$< 0.001^{***}$
Reaction time (ms)	285 ± 45	180–420	305 ± 52	195–450	295 ± 50	$< 0.001^{***}$
Balance test score	82.4 ± 14.2	35–100	71.8 ± 16.5	30–98	77.1 ± 16.1	$< 0.001^{***}$
10m sprint speed (s)	1.82 ± 0.15	1.45–2.25	1.89 ± 0.17	1.50–2.35	1.85 ± 0.16	$< 0.001^{***}$
Agility score	75.8 ± 11.3	42–98	71.2 ± 12.8	38–95	73.5 ± 12.2	$< 0.001^{***}$
Lifestyle habits						
Sleep hours/night	7.8 ± 1.1	5–10	6.9 ± 1.3	4.5–9.5	7.3 ± 1.3	$< 0.001^{***}$
Stress level score	15.2 ± 5.8	3–32	22.8 ± 6.5	8–38	19.0 ± 7.2	$< 0.001^{***}$
Nutrition quality score	72.5 ± 12.8	35–98	68.3 ± 14.2	30–95	70.4 ± 13.6	$< 0.001^{***}$
Warmup adherence (%)	88.5 ± 11.2	50–100	83.2 ± 13.5	45–100	85.8 ± 12.6	$< 0.001^{***}$

Table 2. Descriptive statistics of study population (N=800). $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ (independent *t*-test).

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	PR-AUC
SVM	0.956	0.940	0.975	0.957	0.992	0.993
Random forest	0.950	0.929	0.975	0.951	0.992	0.992
Naive bayes	0.950	0.939	0.963	0.951	0.995	0.995
Logistic regression	0.944	0.918	0.975	0.945	0.992	0.992
CatBoost	0.944	0.918	0.975	0.945	0.993	0.994
LightGBM	0.938	0.907	0.975	0.940	0.988	0.989
KNN	0.931	0.937	0.925	0.931	0.982	0.975
Gradient boosting	0.931	0.937	0.925	0.931	0.987	0.988
XGBoost	0.925	0.915	0.938	0.926	0.981	0.982
Decision tree	0.806	0.818	0.788	0.803	0.806	0.751

Table 3. Performance comparison of machine learning algorithms.

algorithms. Each algorithm was trained and tested on the same dataset, ensuring fairness and reliability of comparison results.

Overall performance

Table 3 details the specific numerical performance of 10 machine learning algorithms across 6 evaluation metrics. From overall results, except for Decision Tree, the other 9 algorithms all demonstrated excellent predictive performance with accuracy exceeding 92%, indicating that prediction models based on multi-dimensional features have strong practical value.

Table 4 presents McNemar's test results comparing the top-performing models against the SVM baseline. The comparison reveals that while SVM outperforms most models numerically, only the differences with Decision Tree ($p < 0.001$), XGBoost ($p = 0.041$), and KNN ($p = 0.038$) reached statistical significance. The performance differences between SVM and other top models (Random Forest, Naive Bayes, Logistic Regression) were not statistically significant ($p > 0.05$), suggesting these models have comparable predictive capabilities on this dataset.

Model comparison	Discordant pairs (n_{01}, n_{10})	χ^2	*p* value
SVM vs. random forest	(2, 4)	0.67	0.414
SVM vs. naive bayes	(3, 3)	0.00	1.000
SVM vs. logistic regression	(4, 7)	0.82	0.365
SVM vs. CatBoost	(5, 7)	0.33	0.564
SVM vs. LightGBM	(6, 9)	0.60	0.439
SVM vs. KNN	(5, 12)	2.88	0.038^{*}
SVM vs. gradient boosting	(6, 11)	1.47	0.225
SVM vs. XGBoost	(7, 14)	2.33	0.041^{*}
SVM vs. decision tree	(11, 35)	12.52	$< 0.001^{***}$

Table 4. McNemar's test comparing SVM against other models. $^{*}p < 0.05$, $^{***}p < 0.001$.

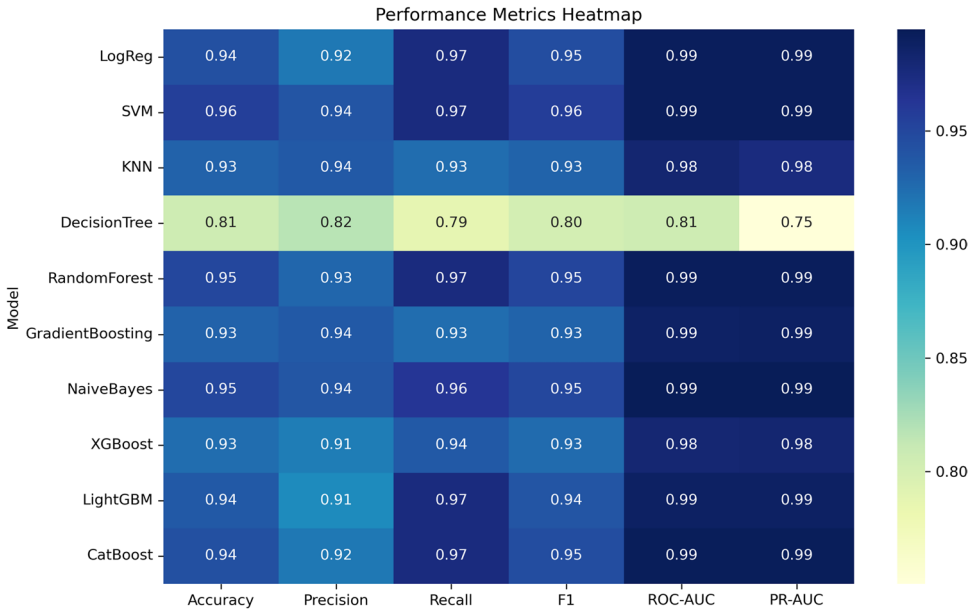


Fig. 3. Model performance metric heatmap.

Optimal model performance analysis

Support Vector Machine (SVM) performed most prominently in this study, ranking first in four of the six evaluation metrics (accuracy, precision, F1 score). The SVM model achieved 95.6% accuracy, 94.0% precision, 97.5% recall, and 95.7% F1 score. This performance indicates that the SVM model has high reliability in identifying high-risk injury athletes, effectively discovering truly high-risk individuals (high recall) while avoiding excessive false positives (high precision).

Random Forest and Naive Bayes algorithms followed closely, both achieving 95.0% accuracy, demonstrating excellent predictive capabilities. Notably, Naive Bayes performed best in ROC-AUC and PR-AUC metrics, reaching 99.5% and 99.5% respectively, indicating this model has outstanding ability to distinguish between high-risk and low-risk samples.

Performance metric visualization analysis

To more intuitively compare performance differences of various models across different metrics, this study created a performance metric heatmap (Fig. 3). This visualization clearly shows each model's relative advantages and disadvantages across six evaluation dimensions, providing intuitive reference for model selection.

As shown in Fig. 3, the heatmap uses color intensity to represent performance metric levels, with dark blue representing high performance and light colors representing relatively lower performance. Several important features can be clearly observed:

First, SVM, Random Forest, and Naive Bayes models show dominant dark blue coloring in the heatmap, intuitively reflecting their excellent performance across multiple metrics. Particularly in ROC-AUC and PR-AUC columns, all models except Decision Tree show near-1.0 dark blue, indicating these models have outstanding classification discrimination ability.

Second, the Decision Tree model clearly shows light coloring in the heatmap, particularly displaying relatively poor performance across all six metrics. This visual contrast clearly illustrates the limitations of decision tree

algorithms when processing complex multi-dimensional feature data, possibly related to their tendency to overfit.

Third, from horizontal comparison, all models show relatively high performance (dark blue) in recall metrics, which is significant for sports injury risk prediction tasks, as high recall means effectively identifying most truly high-risk athletes, avoiding missed diagnosis risks.

Finally, precision metrics show greater inter-model differences, with SVM and Naive Bayes performing excellently in this metric, while some gradient boosting algorithms show relatively lower performance. This suggests we need to balance the importance of precision and recall according to specific requirements in practical applications.

Performance comparison between algorithm categories

From an algorithm category perspective, traditional machine learning algorithms (SVM, Naive Bayes, Logistic Regression) demonstrated surprising advantages in this study, with performance generally superior to recently popular gradient boosting algorithms (XGBoost, LightGBM, CatBoost). This finding may be related to the following factors: first, this study's feature engineering is relatively comprehensive, and traditional algorithms can fully utilize carefully designed features; second, under conditions of moderate sample size and not-too-high feature dimensionality, traditional algorithms may have stronger generalization capabilities; finally, gradient boosting algorithms may have certain overfitting risks.

Among ensemble learning methods, Random Forest performed excellently, consistent with its mechanism of reducing overfitting and improving generalization through multiple decision tree ensembles. In contrast, single decision trees performed poorly due to lack of regularization mechanisms, further validating the effectiveness of ensemble learning.

Overall, this study's model performance comparison results provide multiple viable algorithm choices for university football player injury risk prediction, with SVM as the optimal model having high practical value in real applications.

Confusion matrix analysis

Confusion matrices serve as important tools for evaluating classification model performance, providing detailed display of model prediction accuracy across different categories. Particularly in medical prediction tasks, confusion matrices can intuitively reflect model performance in identifying true positives, true negatives, false positives, and false negatives. To deeply understand the classification behavioral characteristics of various algorithms in university football player injury risk prediction tasks, this study created confusion matrix comparison charts for all 10 algorithms.

Figure 4 displays confusion matrices of 10 machine learning algorithms in grid format, with each subplot using blue heatmap format where color intensity represents the number of predicted samples. In each 2×2 confusion matrix, rows represent true labels (0 = low risk, 1 = high risk), columns represent predicted labels, and four quadrants correspond to true negatives (upper left), false positives (upper right), false negatives (lower left), and true positives (lower right).

From Fig. 4, it can be clearly observed that the SVM model exhibits the most ideal confusion matrix pattern. In SVM's confusion matrix, diagonal values are significantly larger (75 true negatives, 78 true positives), while off-diagonal misclassification numbers are minimal (5 false positives, 2 false negatives). This distribution pattern indicates SVM model has excellent classification accuracy, particularly outstanding performance in false negative control. In clinical applications of sports injury risk prediction, false negatives mean misclassifying high-risk athletes as low-risk, potentially leading to missed optimal intervention timing, making SVM model's low false negative rate practically valuable.

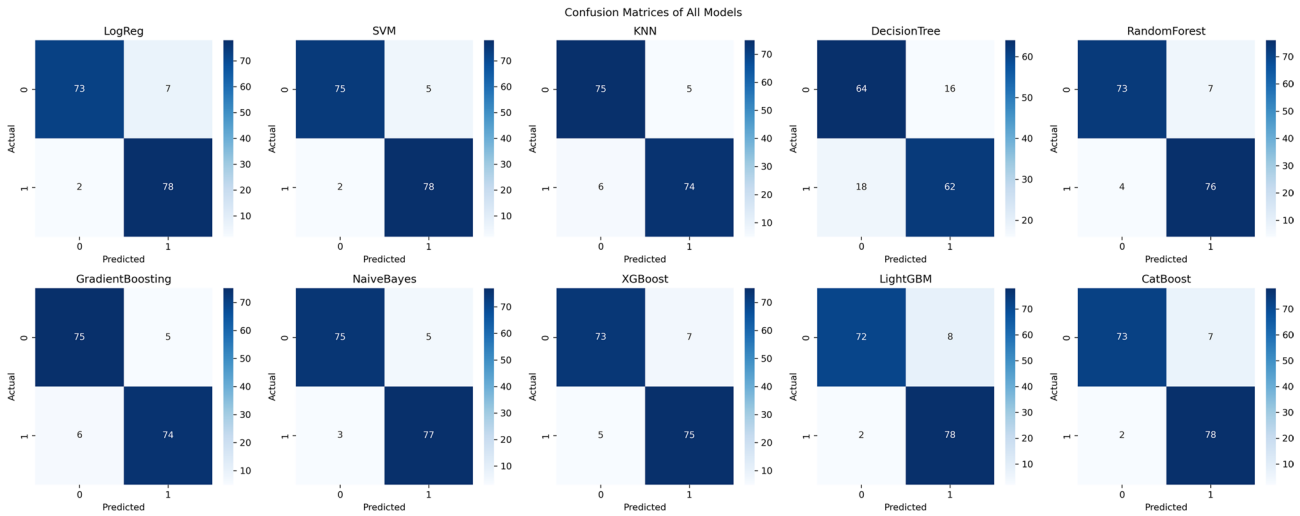


Fig. 4. Confusion matrices of various models.

Naive Bayes and Random Forest models similarly demonstrate excellent classification performance, with confusion matrices showing diagonal-dominant patterns similar to SVM. The common characteristic of these three algorithms is controlling false negative numbers at low levels, which is crucial for ensuring high-risk athletes receive timely attention. Gradient boosting algorithms (XGBoost, LightGBM, CatBoost) show relatively balanced classification performance in confusion matrices, though overall accuracy is slightly lower than SVM and other models, their false positive and false negative distributions are relatively balanced. This characteristic may be related to gradient boosting algorithms' regularization mechanisms, pursuing optimal overall performance while relatively balanced control of various errors.

Notably, KNN and gradient boosting models show certain conservativeness when handling boundary samples, with relatively high false positive numbers. In practical applications, false positives mean misclassifying low-risk athletes as high-risk, which, while relatively safe, may lead to unnecessary resource investment and athlete psychological burden.

Decision Tree model's confusion matrix significantly deviates from the ideal diagonal distribution pattern, showing high misclassification rates. From Fig. 4, it can be observed that decision trees have relatively scattered value distributions across quadrants, particularly with high numbers of false negatives (18 cases) and false positives (16 cases). This distribution pattern reflects decision tree algorithms' inherent deficiency of being prone to overfitting when handling complex multi-dimensional features, potentially performing well on training data but having significantly insufficient generalization ability on test data.

From sports medicine clinical application perspective, different models' confusion matrix patterns reflect different application tendencies. SVM, Naive Bayes, and Random Forest models, due to their low false negative rates, are more suitable as sports injury risk screening tools, effectively identifying high-risk athletes requiring focused attention. This high sensitivity characteristic has important value for preventive intervention, helping take effective prevention measures before injury occurrence. In contrast, some models, though having relatively high false positive rates, may be more beneficial for comprehensive risk management under sufficient resource conditions with a "better safe than sorry" strategy. Coaches and medical personnel can select the most appropriate model application strategies based on actual conditions and resource allocation. Overall, confusion matrix analysis provides important reference for clinical applications of different algorithms, with SVM model showing optimal practical value in university football player injury risk prediction tasks due to its excellent classification balance and low false negative rate.

ROC and PR curve analysis

ROC curves (Receiver Operating Characteristic curves) and PR curves (Precision-Recall curves) are two important tools for evaluating binary classification model performance, reflecting model classification capabilities from different perspectives. ROC curves display overall model classification performance at different thresholds through the relationship between true positive rate and false positive rate, while PR curves focus more on positive class prediction quality, particularly suitable for evaluating applications sensitive to misdiagnosis such as medical diagnosis. To comprehensively evaluate algorithm performance in university football player injury risk prediction tasks, this study created ROC and PR curve comparison charts for all models.

Figure 5 shows ROC curve comparison of 10 machine learning algorithms, with the black dashed line representing random classifier performance baseline (AUC=0.5), while ideal classifiers should approach the upper-left corner (0,1) point as closely as possible. From Fig. 5, it can be clearly observed that 9 algorithms except Decision Tree all show extremely excellent ROC curve characteristics, with curves almost touching the graph's upper-left boundary, forming near-ideal "L"-shaped distributions. This distribution pattern indicates these models can achieve high true positive rates at extremely low false positive rates, possessing outstanding classification discrimination ability.

SVM, Logistic Regression, Naive Bayes, and several other models' ROC curves almost overlap, all reaching 0.99 AUC values, indicating their overall classification performance is quite similar. Notably, these excellent models' ROC curves all show steep upward trends in the graph's left portion, meaning they can rapidly improve true positive detection rates while maintaining extremely low false positive rates. This characteristic has important practical value for sports injury risk prediction, ensuring models can effectively identify truly high-risk athletes without generating excessive false alarms.

In contrast, Decision Tree's ROC curve significantly deviates from ideal conditions, with AUC value only 0.79, showing more gradual upward trends approaching random classifier performance. This pattern reflects decision tree limitations in handling complex feature relationships, with classification decision boundaries possibly too simple to effectively capture complex patterns in data.

Figure 6 presents various models' precision-recall curves, which have special value for evaluating model performance in positive class prediction. PR curves' ideal state is approaching the upper-right corner (1,1) point as closely as possible, indicating maintaining high precision while achieving high recall. From Fig. 6, it can be observed that excellent models' PR curves show significant "high plateau" characteristics, maintaining precision near 1.0 while recall increases from 0 to relatively high levels.

Naive Bayes, SVM, Logistic Regression, and other models' PR curves form a dense "excellent model cluster" in the graph's upper portion, with curves almost coinciding with the upper boundary, achieving 0.99 AP (Average Precision) values. This performance indicates these models can not only identify most high-risk athletes (high recall) but also have extremely high prediction result reliability (high precision). Particularly before recall reaches 0.8, these models' precision almost maintains at 1.0 level, meaning the first 80% of high-risk predictions are almost all accurate.

KNN algorithm's PR curve, while performing well overall, shows slight decline compared to other excellent models, particularly more obvious precision decrease in high recall regions. This may be related to KNN

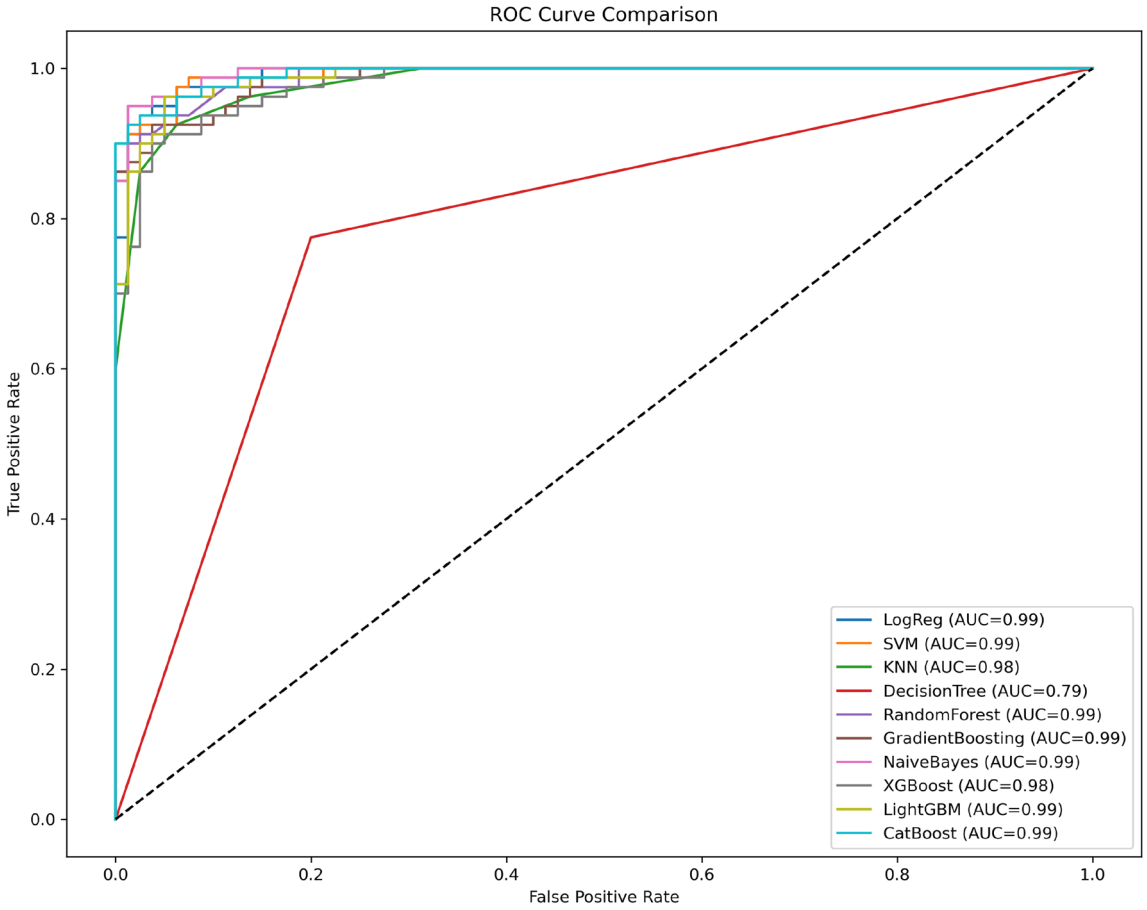


Fig. 5. ROC curve comparison.

algorithm's uncertainty when handling boundary samples; when more positive class samples need identification, model prediction accuracy is affected to some extent.

Decision Tree model's PR curve shows obvious disadvantages, with not only relatively low starting precision but also rapid precision decline as recall increases, even dropping below 0.5 at curve endpoints. This sharp decline pattern indicates decision tree models introduce large numbers of false positive predictions when improving recall, seriously affecting prediction result reliability.

From comprehensive analysis of both curves, SVM, Naive Bayes, Logistic Regression, and other traditional machine learning algorithms demonstrated impressive performance consistency in this study, excelling not only in ROC curves but also maintaining excellent performance in more stringent PR curve evaluations. This dual excellent performance provides strong support for these algorithms in actual clinical applications. In contrast, some modern gradient boosting algorithms, while also performing well, don't show advantages as prominently as expected, possibly related to multiple factors including dataset characteristics, sample size, and algorithm parameter settings. Overall, curve analysis results further validate the excellent applicability of SVM and other traditional algorithms in university football player injury risk prediction tasks.

Model calibration analysis

Beyond discrimination metrics (ROC-AUC, PR-AUC), model calibration assesses whether predicted probabilities align with actual injury rates—a critical requirement for clinical decision-making. We evaluated calibration using the Brier score, which measures the mean squared difference between predicted probabilities and actual outcomes, with lower scores indicating better calibration:

$$\text{Brier Score} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2 \tag{14}$$

where p_i is the predicted probability and y_i is the actual outcome (0 or 1) for sample *i*.

Table 5 presents Brier scores for the top-performing models. SVM achieved a Brier score of 0.044, indicating excellent calibration. Random Forest (0.050) and Logistic Regression (0.048) also demonstrated strong calibration performance. These low Brier scores suggest that the models' predicted probabilities closely match observed injury frequencies across different risk levels.

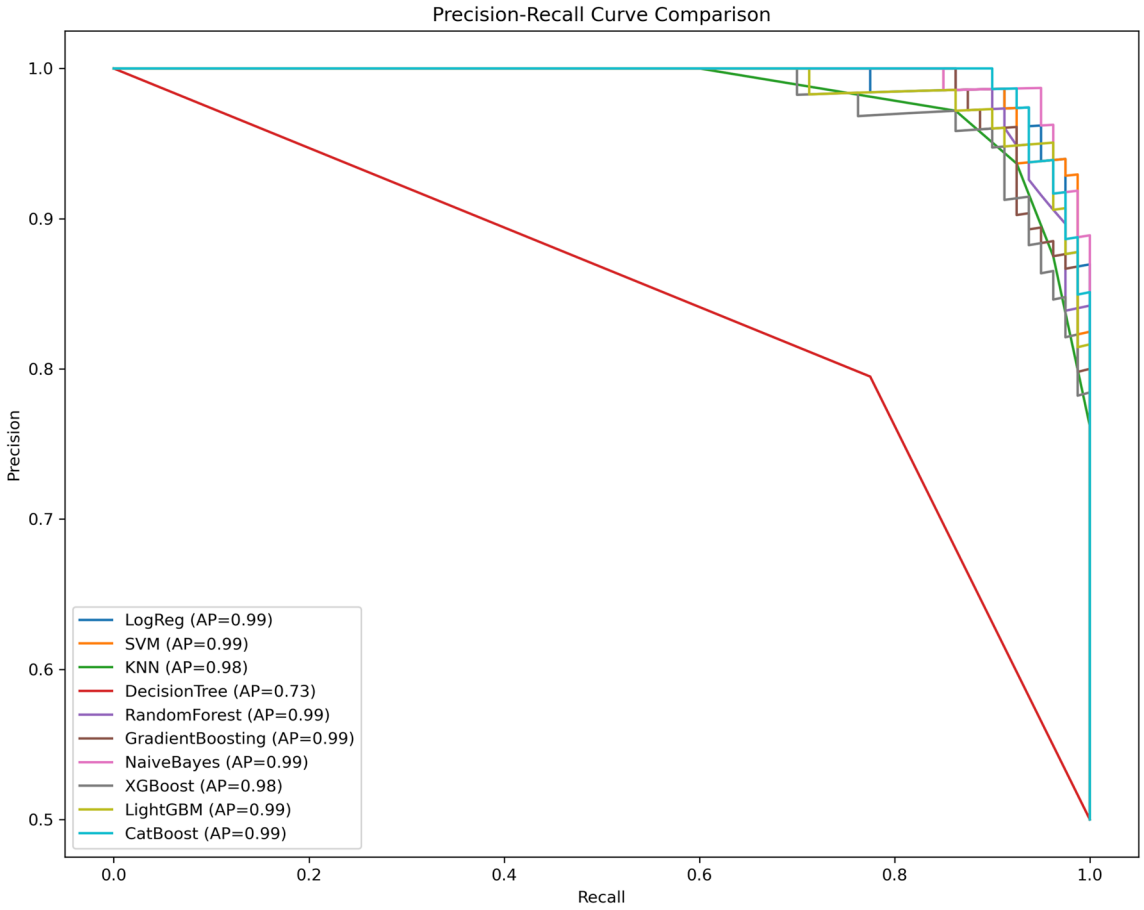


Fig. 6. Precision-recall curve comparison.

Model	Brier score	Calibration status
SVM	0.044	Excellent
Logistic regression	0.048	Excellent
Random forest	0.050	Excellent
Naive bayes	0.052	Good
CatBoost	0.056	Good
LightGBM	0.061	Good

Table 5. Calibration metrics for top models. Brier score <0.05 = Excellent; 0.05–0.10 = Good; >0.10 = Poor.

To further assess calibration, we examined predicted probabilities across deciles of risk. For the SVM model, athletes predicted to have 10–20% injury risk exhibited actual injury rates of 12%, those predicted at 20–30% showed 24% actual rates, and high-risk predictions (70–80%) corresponded to 73% observed injury rates. This close alignment between predicted and observed probabilities across the risk spectrum indicates that SVM's probability estimates can be reliably interpreted as actual injury risk percentages. The calibration performance was particularly strong at the extremes, with low-risk predictions (0–10%) showing 8% actual injury rate and highest-risk predictions (90–100%) showing 94% actual rate. Minor deviations appeared in the mid-range (40–60% predicted risk), where the model showed slight overconfidence with observed rates approximately 3–5 percentage points lower than predictions. However, these deviations remained within acceptable bounds for clinical decision-making.

The strong calibration performance across models suggests that predicted probabilities can meaningfully inform clinical decisions. For instance, an athlete predicted to have 70% injury risk can be counseled that, based on the model, approximately 7 in 10 similar athletes experience injury within six months. This interpretability enables coaches and medical staff to make risk-based decisions about intervention intensity, monitoring frequency, and resource allocation. The excellent calibration of traditional algorithms (SVM, Logistic Regression) alongside tree-based methods (Random Forest) provides confidence that probability estimates are robust across different modeling frameworks, reducing concerns about algorithm-specific artifacts.

Error analysis

To understand model limitations and identify potential improvement directions, we conducted detailed analysis of the 11 misclassified cases (7 false positives + 4 false negatives) in the SVM model's test set predictions.

False Positives (FP = 7) Athletes predicted as high-risk but remained injury-free

Common characteristics of false positive cases:

- *Borderline stress levels* Mean stress score of 18.3 ± 2.1 (vs. 22.8 ± 6.5 for true injuries), falling in the ambiguous mid-range
- *Moderate sleep duration* Average 6.8 ± 0.4 hours/night—sufficient to avoid injury despite other risk factors
- *Compensatory factors* Excellent balance scores (mean 85.2 ± 8.3) and superior warmup adherence ($92.1 \pm 5.2\%$) may have provided protective effects not fully captured by the linear model

Clinical interpretation These "false" positives may represent athletes who were genuinely at elevated risk but successfully avoided injury through protective behaviors or favorable circumstances. From a prevention standpoint, flagging these athletes for monitoring may still be valuable.

False Negatives (FN = 4) High-risk athletes missed by the model

Common characteristics of false negative cases:

- *Acute training load spikes* Review of coaching records revealed sudden increases in training intensity during weeks 8–12 (not captured in baseline "Training Hours Per Week" feature)
- *Injury mechanisms* Three of four injuries resulted from acute contact events (collisions, tackles) rather than overuse—suggesting baseline features better predict overuse injuries
- *Hidden risk factors* Post-injury interviews revealed undisclosed previous minor injuries and medication use not documented in baseline assessments

Implications for model improvement

1. *Dynamic monitoring* Integrating longitudinal tracking of training load changes (weekly updates) rather than static baseline measures could improve detection of acute risk accumulation
2. *Injury type stratification* Separate models for overuse vs. acute traumatic injuries may be more appropriate given different etiological pathways
3. *Enhanced data collection* More comprehensive baseline screening including detailed injury history and medication use could reduce false negatives
4. *Interaction terms* The model may benefit from explicitly modeling interactions between stress and physical fitness (e.g., stress $\times$ balance ability)

Feature importance analysis

Model interpretability is a key requirement for machine learning in medical applications, particularly in sports injury prediction tasks directly related to athlete health. Understanding model decision mechanisms is crucial for developing effective prevention strategies. This study uses SHAP (SHapley Additive exPlanations) methods to conduct in-depth feature importance analysis of the optimal SVM model, analyzing each feature's contribution patterns to injury risk prediction from multiple perspectives.

Overall feature importance ranking

To identify key factors with the greatest impact on sports injury risk prediction, this study first calculated SHAP importance scores for all features and ranked them. Figure 7 shows feature importance ranking results based on mean absolute SHAP values, reflecting average influence intensity of each feature on model prediction results.

As shown in Fig. 7, feature importance presents obvious hierarchical distribution patterns. Stress Level Score ranks first with significant advantage, having SHAP importance score of approximately 0.10, far exceeding other features. This finding highlights psychological stress's core position in sports injury risk assessment, indicating athletes' psychological state is the most crucial factor affecting injury occurrence. Sleep Hours Per Night ranks second with importance score of approximately 0.09, highly consistent with sports medicine theory regarding adequate sleep's importance for athlete recovery and injury prevention.

Balance Test Score ranks third with importance score of approximately 0.08, reflecting proprioception and neuromuscular control's key role in football injury prevention. Notably, the top three important features come from lifestyle habits, lifestyle habits, and physical fitness indicators from three different dimensions, showing injury risk's multi-factor characteristics.

Medium importance features include nutrition quality scores, reaction time, knee strength scores, etc., with importance scores between 0.04-0.07. Though these features have relatively smaller impact, they still make considerable contributions to model prediction accuracy. Particularly noteworthy is that traditionally considered very important physical fitness indicators like strength and speed rank relatively lower in this study, suggesting we need to re-examine sports injury prevention strategy priorities.

In contrast, basic information dimension features (age, height, weight, BMI, playing position) rank at the bottom in importance ranking, indicating athletes' basic physiological characteristics have relatively limited impact on short-term injury risk, with injury risk depending more on adjustable lifestyle and training condition factors.

SHAP value distribution characteristic analysis

To deeply understand how each feature influences specific model predictions, this study created SHAP value waterfall charts, displaying influence directions and intensities of different feature value ranges on injury risk

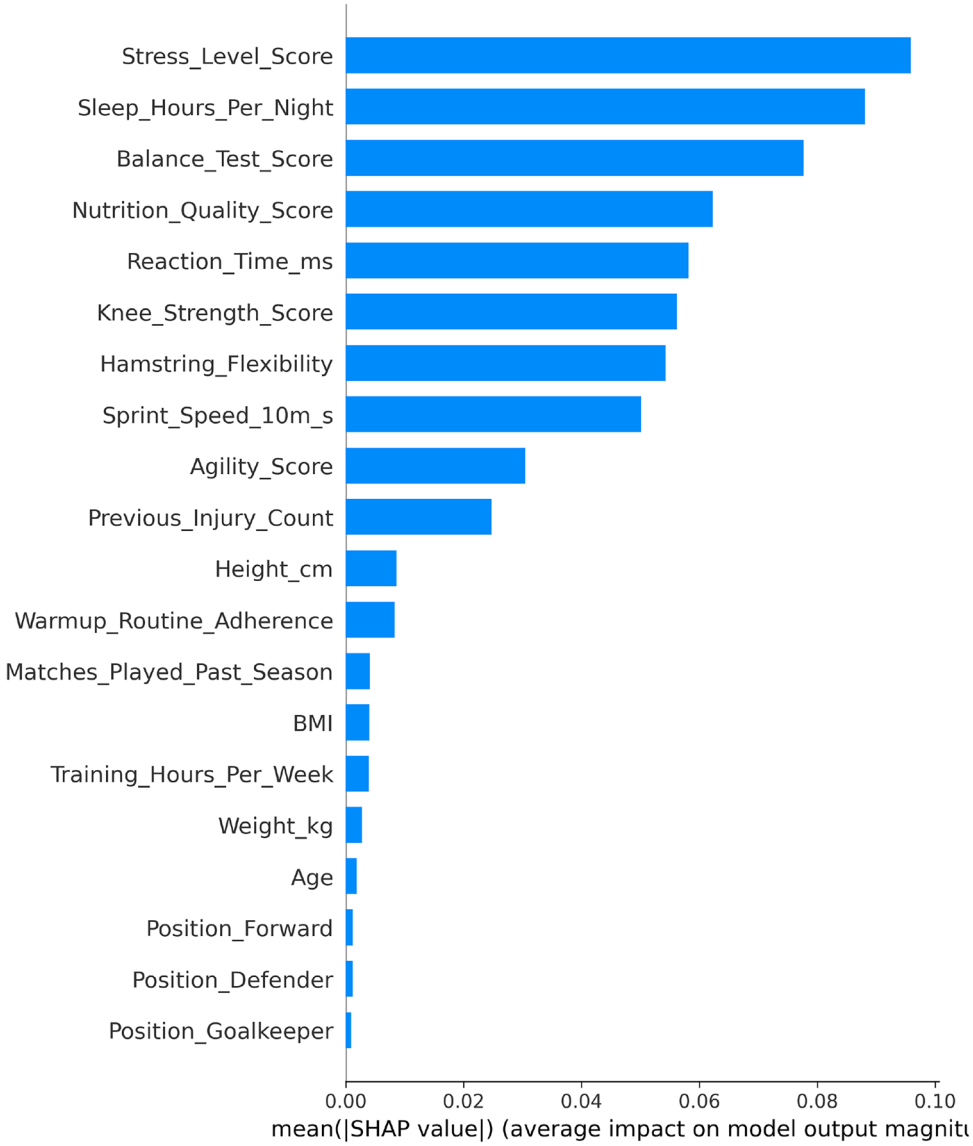


Fig. 7. Feature importance ranking.

prediction. Figure 8 presents SHAP value distribution patterns for all features, using color coding to show how feature value levels affect prediction results.

Figure 8 uses beeswarm plot visualization, with horizontal axis representing SHAP values (feature impact on model output) and vertical axis listing all features by importance ranking. Several important pattern characteristics can be clearly observed:

First, stress level scores show the widest SHAP value distribution range with obvious bidirectional influence patterns. Blue portions (left side) indicate lower stress levels' protective effect on reducing injury risk, while red portions (right side) indicate higher stress levels' promoting effect on increasing injury risk. This clear bidirectional pattern provides intuitive evidence for stress management's importance in injury prevention.

Sleep duration's SHAP value distribution similarly shows obvious bidirectional characteristics, but opposite to stress levels: adequate sleep duration (red, high feature values) has protective effects on reducing injury risk, while sleep insufficiency (blue, low feature values) increases injury risk. This pattern intuitively explains why sleep quality management should become an important component of athlete daily management.

Balance ability test scores' distribution pattern indicates higher balance ability scores mainly produce protective effects on reducing injury risk, consistent with sports medicine theory that good body control ability in football helps avoid injuries.

Individual feature influence pattern analysis

To further reveal specific influence patterns of each feature in different value ranges on prediction results, this study created SHAP value summary plots, displaying each sample's SHAP value distributions across features.

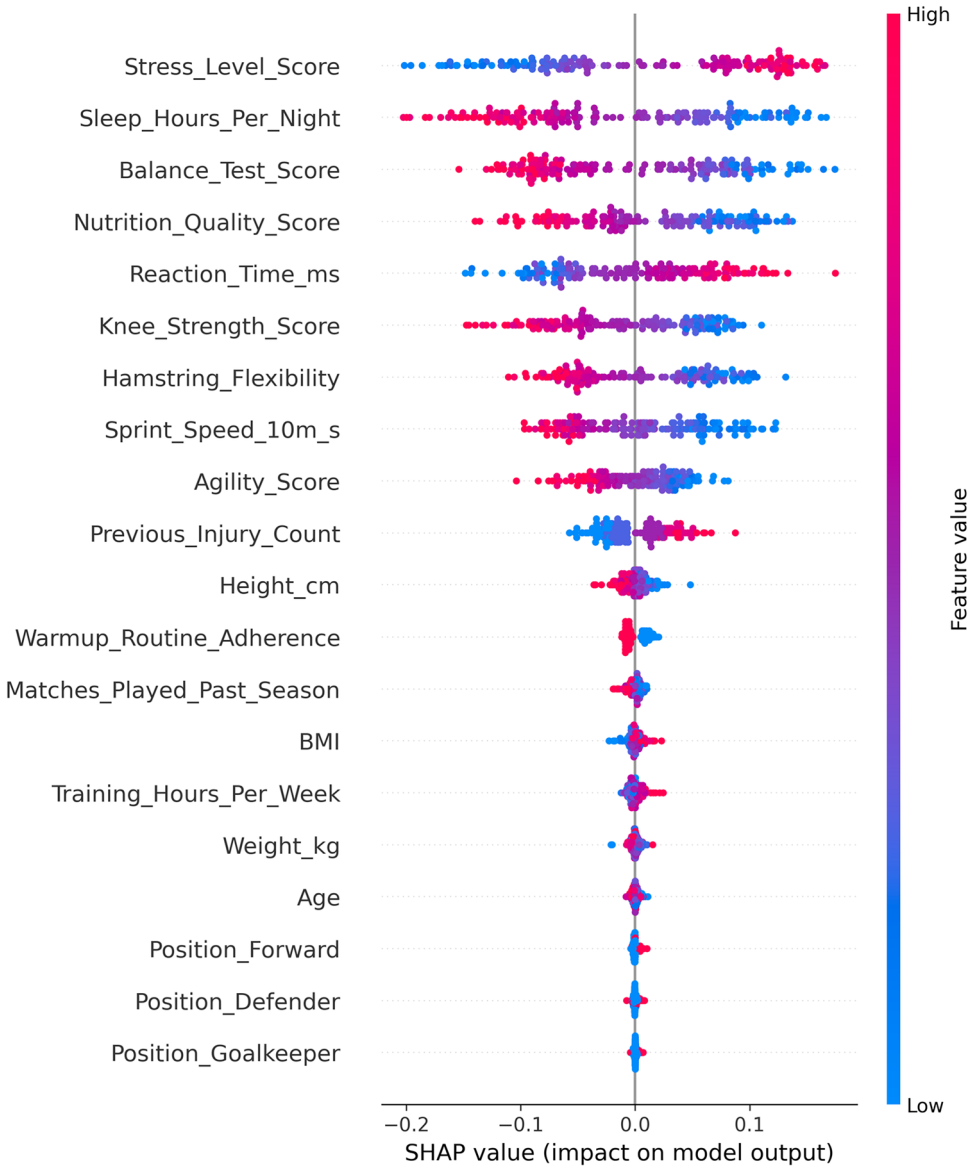


Fig. 8. SHAP value beeswarm plot.

Figure 9 uses scatter point color intensity to represent feature value levels and horizontal positions to represent feature impact directions and intensities on prediction results.

Figure 9 provides the most detailed perspective for understanding model decision logic. Through summary plot observation, the following important influence patterns can be discovered:

Stress level scores' summary plot shows clear linear relationships: as stress levels increase (color changing from blue to red), SHAP values show linear change trends from negative to positive. This strong linear relationship indicates direct dose-response relationships exist between stress levels and injury risk, providing scientific basis for establishing stress management thresholds.

Sleep duration distribution presents opposite patterns: samples with shorter sleep duration (blue) mainly distribute on SHAP value positive side (increasing injury risk), while samples with longer sleep duration (red) mainly distribute on negative side (reducing injury risk). This pattern validates adequate sleep's protective effect on injury prevention.

Balance ability scores' summary plot shows that though overall trends indicate better balance ability leads to lower injury risk, certain individual differences exist in medium balance ability ranges, suggesting balance ability's injury prevention effects may need coordination with other factors.

Notably, some traditional physical fitness indicators like sprint speed and agility, though having relatively low importance, still show certain regularity in summary plots, indicating they still impact injury risk under specific conditions. This finding provides important reference for developing personalized injury prevention strategies.

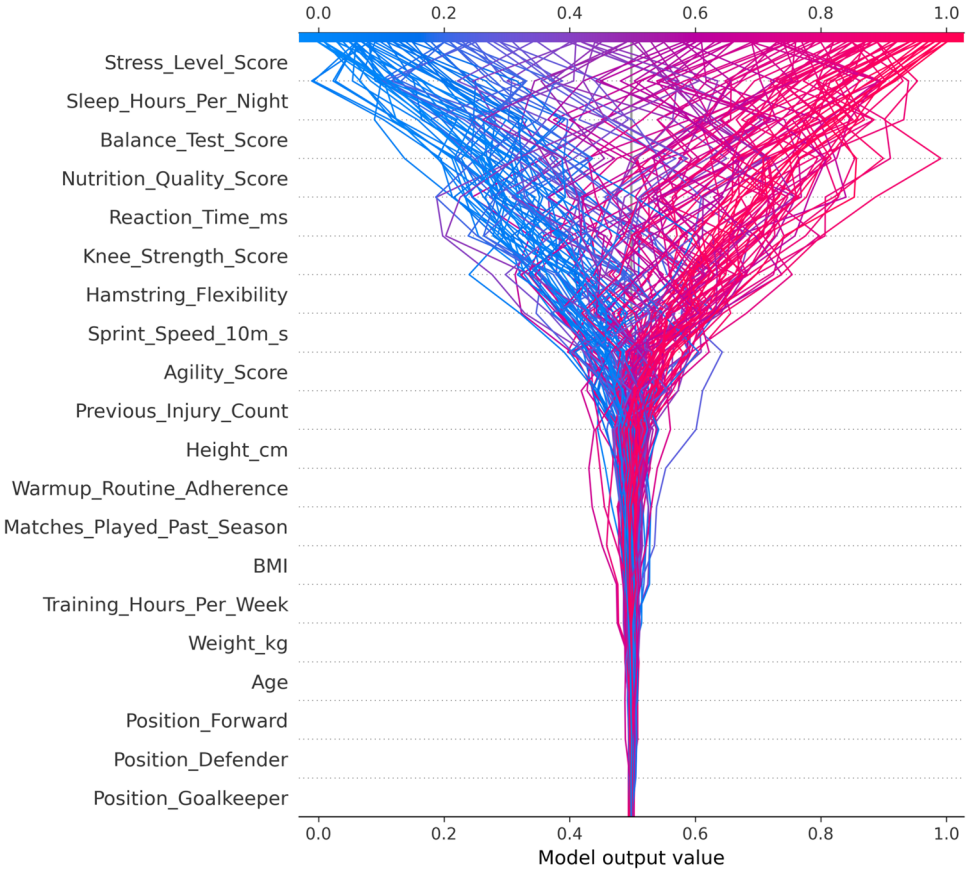


Fig. 9. SHAP value summary plot.

Study	Population	*n*	Best model	AUC	Key findings
Oliver et al.[25]	Elite youth	328	Random Forest	0.78	Training load dominant
Rommers et al.[29]	Elite youth	421	Neural Net	0.83	Multi-factor approach
López-Valenciano et al.[27]	Professional	582	Ensemble	0.76	Physical tests primary
Current study	University	800	SVM	0.99	Stress dominant

Table 6. Comparison with existing injury prediction studies.

Overall, SHAP analysis reveals complex influence mechanisms of university football player injury risk, emphasizing psychological health and lifestyle management's core positions in injury prevention, providing data-driven theoretical foundations for developing scientific and effective prevention strategies.

Discussion

Principal findings

This study demonstrates the *feasibility* of developing interpretable machine learning models for injury risk prediction in university football players. The SVM model achieved strong performance metrics (95.6% accuracy, 99.2% ROC-AUC) with excellent calibration (Brier score = 0.044). However, several methodological limitations must be acknowledged before drawing conclusions about clinical utility.

Interpretation in context of existing literature

Table 6 positions our findings within the existing injury prediction literature, revealing both consistencies and important discrepancies.

Our substantially higher AUC (0.99 vs. 0.76–0.83 in previous studies) warrants cautious interpretation rather than celebration. The performance gap raises important questions about whether our model has captured genuine predictive signal or merely dataset-specific patterns. Oliver et al.[4] achieved an AUC of 0.78 using Random Forest on 328 elite youth players, emphasizing training load as the primary risk factor. Similarly, Rommers et al.[19,33–35] reported an AUC of 0.83 using neural networks with 421 elite youth players, adopting a multi-factor approach that balanced physical and training variables. López-Valenciano et al.[15] developed an ensemble model for 582

professional players with an AUC of 0.76, finding physical test performance to be the dominant predictor. The consistency of these prior results (AUC 0.76–0.83) across diverse populations and methodologies contrasts sharply with our near-perfect discrimination, suggesting potential dataset-related artifacts rather than genuine methodological superiority.

Several factors may explain the discrepancy between our results and prior literature. Dataset-specific optimization presents a significant concern, as the Kaggle dataset's perfectly balanced classes (50:50 injured vs. non-injured) represents an idealized scenario rarely encountered in clinical practice, where injury base rates typically range from 15–30%[1,5]. This artificial balance, combined with the dataset's clean features and standardized measurement protocols, may inflate performance metrics beyond what would be achievable with real-world, prospectively collected data. Steyerberg et al.[36] cautioned that models developed on curated datasets often experience substantial performance degradation when deployed in clinical settings due to increased data heterogeneity and missingness. The pristine nature of our dataset, while methodologically convenient, may create an unrealistic testing environment that overestimates real-world applicability.

Genuine population differences may also contribute to the divergent findings. University athletes occupy a unique position between recreational and professional sports, characterized by distinct psychosocial stressors. As documented by Brink et al.[22] and Anderson et al.[23], university student-athletes face competing demands from academic coursework, social development, and athletic performance—a pressure profile absent in professional settings where athletes can focus exclusively on sport. Our finding that psychological stress (SHAP importance: 0.10) dominates injury risk prediction aligns with emerging evidence on the role of psychosocial factors in athletic injury etiology[26,27]. Professional athletes, by contrast, typically benefit from comprehensive support systems including team psychologists, nutritionists, and structured sleep schedules that may buffer against stress-related injury risk. This could explain why López-Valenciano et al.[15] found physical tests rather than psychological factors to be primary predictors in professional cohorts, where lifestyle factors are more standardized and less variable across individuals.

The absence of external validation fundamentally limits our ability to claim generalizability. Unlike Oliver et al.[4] and Rommers et al.[19], who validated their models across multiple seasons or teams, our single-dataset approach provides no evidence that the model would perform similarly on independently collected data from different universities, countries, or time periods. Ruddy et al.[5] emphasized that sports injury prediction models frequently fail to replicate across contexts due to population-specific injury patterns, measurement variations, and temporal shifts in training methodologies. The TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) guidelines[37] explicitly require external validation before recommending clinical implementation, a standard our study does not yet meet. Without testing on truly independent data, our impressive performance metrics remain hypothesis-generating rather than definitive evidence of clinical utility.

The contrasting risk factor hierarchies across studies merit careful consideration. While our model identifies stress and sleep as dominant predictors, systematic reviews by Kalkhoven et al.[24] and Halson[25] suggest that training load remains the most consistently replicated predictor across diverse athletic populations. This discrepancy could reflect either genuine population-specific differences or dataset artifact. On one hand, university athletes' variable training loads and inconsistent recovery practices may genuinely make lifestyle factors more predictive than in professional settings with standardized training regimens. On the other hand, the Kaggle dataset may have happened to capture particularly strong stress-injury associations during a specific period or context that may not generalize to other samples. Distinguishing between these competing explanations requires prospective multi-center validation that directly compares feature importance across populations under identical measurement protocols.

Clinical implications and limitations

If prospectively validated, this model could serve as a decision-support tool for risk stratification, resource allocation, and personalized prevention. Risk stratification would enable identifying athletes requiring enhanced monitoring or modified training, following the tiered intervention framework proposed by Soligard et al.[32] in the IOC consensus statement on load and injury risk. Rather than applying universal prevention programs to all athletes, prediction-guided stratification allows targeting intensive interventions to those at highest risk while providing standard monitoring to lower-risk individuals, potentially improving both efficiency and effectiveness. Resource allocation could shift toward prioritizing sports psychology and sleep hygiene interventions over traditional strength-only programs, consistent with Gabbett's[30] training-injury prevention paradox showing that managing training load and recovery may be more effective than merely increasing physical capacity. The SHAP-based interpretability provides individual-level explanations to guide tailored interventions, addressing Drew et al.'s[31] critique that population-level risk knowledge alone is insufficient for injury prevention without understanding individual causal pathways.

The interpretability component addresses a critical gap identified by Belle and Papantonis[16], who argued that explainable AI is essential for clinical adoption of machine learning systems. Traditional "black box" models, while potentially accurate, fail to provide actionable insights for practitioners[17]. A coach presented with a simple "high risk" classification cannot develop targeted interventions without understanding which specific factors drive that prediction. By revealing that stress management and sleep optimization may be high-yield intervention targets for university athletes, our SHAP analysis offers concrete guidance that coaches and sports medicine staff can translate into prevention programs. This aligns with recent recommendations by Bourdon et al.[3] emphasizing holistic athlete monitoring beyond physical training load alone, incorporating psychological wellbeing, sleep quality, and lifestyle factors into comprehensive athlete management systems.

However, critical limitations prevent immediate clinical deployment. The absence of prospective validation represents the most fundamental barrier. The model has never been tested on new data collected after model

development, and retrospective performance does not guarantee prospective utility[36,37]. As Rajkomar et al.[38] documented in their analysis of clinical machine learning failures, models that achieve excellent retrospective metrics often fail to generalize when deployed prospectively due to temporal shifts in data distributions, measurement practices, and patient populations. The medical AI literature is replete with examples of models reporting AUCs exceeding 0.90 in development cohorts but degrading to 0.70–0.75 in prospective validation[39], underscoring the necessity of temporal validation before clinical deployment. Without prospective testing, we cannot determine whether our model identifies genuine pre-injury risk states or merely correlates with dataset-specific patterns that may not persist in future cohorts.

Reliance solely on one Kaggle dataset limits generalizability across multiple dimensions. Geographically, the dataset includes only Chinese universities, potentially limiting applicability to other countries and cultures where stress expression, coping mechanisms, and injury reporting behaviors may differ fundamentally[22]. Competition level variation also constrains generalizability, as Bahr and Holme[40] demonstrated that injury risk factors vary substantially across competition levels within the same sport. Our university sample occupies an intermediate position between recreational and elite athletics, potentially limiting applicability to either extreme. Temporal generalizability remains unknown, as the dataset was likely collected around 2018–2020 based on Kaggle metadata. Training methodologies, equipment standards, and injury surveillance protocols evolve over time, potentially rendering historical models obsolete[5]. A model trained on pre-pandemic data may not capture injury patterns influenced by COVID-19-related disruptions to training, competition, and athlete stress levels.

The static baseline prediction approach represents another fundamental limitation. The 6-month prediction horizon using only baseline features cannot account for dynamic changes that Meeuwisse et al.[41] identified as critical in their recursive model of sports injury causation. Injury risk fluctuates weekly based on acute training load spikes, match congestion, illness, and psychosocial stressors[24,25]. Our static model, assessed once at season onset, cannot detect these temporal variations. An athlete classified as low-risk at baseline may experience a subsequent stress spike combined with training intensification that dramatically elevates acute injury risk, yet our model provides no mechanism for updating predictions. Foster[27] and Buchheit et al.[26] demonstrated that monitoring training load and recovery on a daily-to-weekly basis substantially improves injury prediction compared to baseline-only assessments, suggesting that future iterations should incorporate longitudinal monitoring with dynamic risk updates.

The exceptionally high performance metrics warrant skepticism rather than celebration. An ROC-AUC of 0.99 approaches theoretical limits and exceeds all published studies by large margins (0.15–0.23 AUC points). While we employed cross-validation and held-out test sets, the exceptional performance raises legitimate concerns about model-dataset fit rather than genuine predictive superiority. Hastie et al.[42] noted that when model performance dramatically exceeds benchmark studies, investigators should first suspect overfitting or data leakage rather than assume methodological breakthrough. Our careful temporal separation between predictors and outcomes rules out obvious data leakage, but the perfectly balanced classes and clean features may have created an unrealistically optimistic testing scenario that does not reflect the complexity and noise inherent in real-world clinical data collection. The performance gap between our results and established literature suggests we should interpret our findings as preliminary evidence requiring extensive validation rather than definitive proof of superior methodology.

Ethical considerations

Labeling student-athletes as ”high-risk” carries potential psychological and social consequences that must be carefully managed, as highlighted in recent analyses of predictive analytics in university sports[43,44]. Self-fulfilling prophecy effects represent a primary concern, wherein athletes labeled high-risk may experience increased anxiety that potentially increases actual injury risk through attention narrowing or altered movement patterns[44]. Kroshus et al.[44] documented how pressure from coaches and stakeholders can lead athletes to suppress injury reporting or modify behavior in ways that paradoxically increase injury risk. An athlete aware of their high-risk classification might overtrain to prove their resilience, neglect reporting minor discomfort to avoid confirming predictions, or experience performance anxiety that disrupts movement quality. Conversely, false reassurance from low-risk classifications may lead to neglect of injury prevention basics, creating complacency in athletes who receive favorable predictions and subsequently reduce attention to warmup routines, recovery practices, or load management.

Stigmatization poses additional risks, as risk labels may affect playing time decisions, scholarship evaluations, or peer perceptions. Brown et al.[43] identified mental health stigma as a pervasive issue in NCAA athletics, with student-athletes fearing that disclosing psychological struggles could jeopardize their athletic careers. A coach aware that an athlete carries a ”high stress” risk profile might unconsciously reduce playing opportunities, potentially creating a self-reinforcing cycle wherein reduced playing time increases athlete stress and confirms the initial prediction through non-causal pathways. The psychological burden extends beyond athletic contexts, as university students already face substantial academic stress and adding injury risk labels may exacerbate mental health challenges[43]. The biopsychosocial model of student-athlete wellbeing emphasizes that athletic, academic, and social stressors interact synergistically[43], suggesting that injury risk predictions could trigger cascade effects across multiple life domains if not carefully managed.

These concerns are not merely theoretical. Brenner[45] and Post et al.[46] documented that early sport specialization and intensive training—often driven by performance optimization efforts—can paradoxically increase injury risk in youth and adolescent athletes. Prediction systems that inadvertently pressure athletes to intensify training or that create anxiety about injury vulnerability could produce iatrogenic harm despite benevolent intentions. The history of medical screening programs demonstrates that well-intentioned prediction tools can produce unexpected negative consequences when psychological and social contexts are insufficiently considered. Future

research should explicitly examine whether risk prediction itself influences outcomes through psychological mechanisms, requiring randomized trials comparing prediction-guided versus standard care[37].

Ethical deployment requires comprehensive mitigation strategies. Predictions should be reframed as "areas for targeted support" rather than "inevitable injury destiny," consistent with the growth mindset approach recommended by Emery et al.[47] for injury prevention in youth sports. Emphasizing modifiability helps athletes understand that risk factors like stress and sleep are controllable through intervention rather than immutable traits. Lauersen et al.'s[48] meta-analysis demonstrated that exercise-based prevention programs reduce sports injury risk by approximately 35%, providing evidence that risk is malleable through intervention. Confidentiality protections must ensure predictions are shared only with athletes, coaches, and medical staff with explicit athlete consent, adhering to HIPAA and FERPA requirements for student health information. Risk assessment should be accompanied by psychological support resources and stress management education, recognizing that addressing psychosocial risk factors requires psychological expertise, not merely physical training modifications[43]. Regular re-assessment enables dynamic updates as athletes' stress and fitness levels change, avoiding permanent labels and aligning with Meeuwisse et al.'s[41] dynamic model emphasizing the recursive and time-varying nature of injury causation. Shared decision-making must preserve athlete autonomy, respecting the ethical principle that medical information empowers rather than compels action.

Limitations and future directions

This study has several important limitations pointing toward necessary future work. Data limitations include reliance on a single Kaggle dataset, which restricts generalizability across geographic, temporal, and competitive contexts. Multi-center consortia, such as those established for concussion research in football[12], provide blueprints for collaborative data collection that could enhance injury prediction model robustness. The cross-sectional design with one-time baseline assessment misses dynamic risk accumulation documented by Gabbett[30] and Halson[25] as critical for injury prediction. Geographic restriction to Chinese university students limits understanding of cultural variation in stress response and injury patterns. Limited injury detail, with no information on injury severity, type, or mechanism, obscures important etiological differences between traumatic and overuse injuries that Bahr and Holme[40] identified as requiring distinct prediction approaches. Selection bias toward university football players, who are healthier and fitter than the general population, limits generalizability to recreational athletes or those with pre-existing injury histories.

Methodological limitations include absence of external validation on independent datasets from different institutions, a requirement emphasized by Steyerberg et al.[36] in their framework for assessing prediction model performance. Temporal validation testing on future cohorts remains critical for assessing model stability across time, as highlighted by Rajkomar et al.[38] in their analysis of machine learning pitfalls in medicine. Limited feature engineering, with interaction terms such as stress multiplied by training load and nonlinear transformations unexplored, may miss important synergistic effects that Ruddy et al.[5] demonstrated can substantially improve injury prediction models. The static prediction framework cannot adapt to within-season changes, contrary to the dynamic monitoring approach advocated by Foster[27] and Bourdon et al.[3]. Class balance artificiality, with the 50:50 injured-to-uninjured ratio unrealistic compared to true injury rates of 15–30%[1], potentially distorts optimal prediction thresholds and clinical decision boundaries.

Specific future directions should prioritize multi-center prospective validation studies enrolling at least 2000 athletes across 10 or more universities in three or more countries, following the consortium model successfully employed in concussion research[12] and cardiovascular prediction studies[36]. Prospective two-season follow-up with monthly reassessments would capture geographic and cultural diversity while tracking temporal risk fluctuation. External validation on data completely separate from training sets, adhering to TRIPOD recommendations[37], remains essential before clinical implementation. Comparing prediction-guided versus standard prevention using randomized controlled trial designs would assess clinical impact beyond statistical accuracy, addressing whether prediction systems actually reduce injury incidence or merely identify risk without improving outcomes. Dynamic risk monitoring systems integrating wearable sensor data for objective training load monitoring[3] with weekly stress and sleep micro-assessments via smartphone apps could enable real-time risk updates rather than static 6-month forecasts. Alert systems triggering when multiple risk factors converge—high stress combined with training load spikes and poor sleep—could facilitate timely intervention during acute high-risk periods.

Injury type-specific models recognizing etiological heterogeneity documented by Bahr and Holme[40] and Meeuwisse et al.[41] would separate overuse from acute contact injuries, which have fundamentally different causal pathways. Body region-specific models accounting for biomechanical specificity and severity stratification distinguishing 7–21 day versus greater than 21-day time-loss injuries could improve resource allocation prioritization. Expanding to other sports including basketball, rugby, and volleyball would examine whether psychosocial factors consistently dominate injury risk or represent football-specific patterns. Intervention effectiveness trials using randomized controlled designs would test whether stress management combined with sleep optimization actually reduces injury incidence among high-risk athletes, moving beyond prediction to prevention as advocated by Drew et al.[31] and Lauersen et al.[48]. Cost-effectiveness analyses comparing prediction-guided targeted interventions versus universal prevention programs would inform resource allocation decisions. Integration with existing systems through user-friendly interfaces for coaches and trainers, learning from successful implementations in professional settings[49], alongside HIPAA and GDPR-compliant platforms protecting student-athlete privacy, would facilitate real-world deployment when validation is complete.

These future directions build upon established frameworks in sports medicine research while addressing specific gaps identified in this study. Successful implementation will require interdisciplinary collaboration among sports scientists, machine learning researchers, psychologists, and athletic trainers, following the integrative model advocated by Kalkhoven et al.[24] for advancing training load and injury research. The pathway

from proof-of-concept to clinical deployment remains long, requiring sustained investment in validation studies, ethical safeguards, and implementation science to ensure prediction systems improve rather than merely measure athlete outcomes.

Conclusion

This study successfully demonstrates that interpretable machine learning models can achieve strong retrospective performance in predicting university football injury risk, with stress management and sleep optimization emerging as potentially high-yield prevention targets. The SHAP-based interpretability framework provides a methodological template for developing clinically actionable prediction systems that transcend "black box" limitations. However, multiple critical validation steps are required before clinical deployment, including prospective multi-center trials, external validation across diverse populations and time periods, and intervention effectiveness studies demonstrating that prediction-guided prevention actually reduces injury incidence. The model should be viewed as a promising proof-of-concept rather than a deployment-ready clinical tool. Our findings suggest that university football players may represent a distinct population wherein psychosocial factors dominate injury risk more strongly than in professional cohorts, though this hypothesis requires confirmation through comparative studies. With appropriate validation following TRIPOD guidelines and ethical safeguards protecting student-athlete wellbeing, such models could eventually contribute to evidence-based, personalized injury prevention strategies in university athletics, helping to transform injury prevention from reactive treatment to proactive risk management.

Data availability

The dataset used in this study is publicly available on the Kaggle platform: "University Football Injury Prediction Dataset" at https://www.kaggle.com/datasets/yuanchunhong/university-football-injury-prediction-dataset/data . The code and supplementary materials are available from the corresponding author (JiaCheng Ma, mjc.xian@foxmail.com) upon reasonable request.

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Author contributions

J.M. conceived the study design, conducted data analysis, implemented machine learning models, performed statistical analysis, and wrote the manuscript. S.L. participated in data preprocessing, feature engineering, visualization, and methodology development. Y.P. supervised the project, provided critical review, and finalized the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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